

ATIQRUR RAHAMAN

✉ rhatik.atik.1@gmail.com  <https://www.linkedin.com/in/atiquurrahaman/>

 <https://atiquur1235576689.github.io/>

Education

Bangladesh University of Engineering & Technology (BUET)

B.Sc. in Naval Architecture and Marine Engineering

Dhaka, Bangladesh

Jan. 2022 – Jun. 2026

- CGPA: 3.78 / 4.00 (3rd in the Class)

Research Interest

- Computational Fluid Dynamics
- Battery Thermal Management
- Heat Transfer
- Phase Changing Material
- Fluid Mechanics
- Offshore Structure

Research Experience

• Ongoing Research (Overview)

Aug. 2026 – Present

Project title: *Iso-Volume Comparison of Fin Geometries for PCM-Based Battery Thermal Management: Shape Ranking under Repeated Charge and Discharge Cycling.*

- Ranking fin geometries for PCM-based battery thermal management in **ANSYS Fluent**
- Measuring how long each arrangement holds the cell below its safe operating temperature
- Running many charge and discharge cycles, tracking residual melt fraction between cycles

• Undergraduate Thesis Student, NAME, BUET (Overview)

Apr. 2025 – Jun. 2026

Thesis title: *Numerical Investigation of Mooring Line Angular Spacing on the Dynamic Motion of a 15MW Semi-Submersible Offshore Wind Turbine Platform.*

- Modelled the IEA 15 MW reference turbine on the UMaine VoltturnUS-S platform
- Coupled frequency-domain and time-domain analysis in **ANSYS AQWA**
- Swept mooring line spread angles across a range of wave headings

• Independent Research Collaboration (Overview)

Jul. 2025 – Present

Project title: *Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio.*

- Studied K-shaped fins in a PCM enclosure in **ANSYS Fluent 2024R1**
- Swept a range of fin angles and arm length ratios; intermediate angles with a longer lower arm raised melting efficiency
- Contribution: conceptualization, simulations, formal analysis, data curation, and production of the contour plots, efficiency comparisons, and LMLI curves

Publications

- Dr. Md. Shahidul Islam, **Md. Atiqur Rahaman**, and Md Tanvir Anjum Zenith, “Wave-Heading Dependence of the Mooring Spread-Angle Effect on the Wave-Frequency Motion of a 15 MW Semi-Submersible Floating Offshore Wind Turbine” (*Submitted for Review in Journal of Offshore Mechanics and Arctic Engineering (ASME).*). (Overview)
- Md Tanvir Anjum Zenith, Adri Das, Shafollo Proshad Gupto, **Md. Atiqur Rahaman**, Md. Khairul Islam, “Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio.” (*In preparation*) (Overview)

Professional Experience

Intern (Overview)

Jan. 2024 – Feb. 2025

Dockyard & Engineering Works Ltd, BANGLADESH NAVY – Narayanganj, Bangladesh

- Completed a three-week industrial attachment at a working shipyard
- Trained on marine propulsion systems, shipboard air conditioning, thermal management of engine rooms, and CNC plate cutting

Project title: <i>Design of a 175 Passenger Ferry for Circular Waterways around Dhaka.</i>	<i>2024</i>
• Worked through the design spiral from principal dimensions to the lines plan • Calculated hydrostatics, resistance, and powering in linked Excel sheets • Estimated weight and checked stability against the design requirements • Selected the propeller and ran a cavitation check • Sized the engine, gearbox, and engine foundation	

Technical Skills

Simulation Software	ANSYS Fluent, ANSYS AQWA, ABAQUS
Programming Languages	C++, Python, Fortran
CAD & 3D Modelling	AutoCAD, Rhinoceros 3D
Ship Design Software	MAXSURF, DELFTship
Tools	Word, Excel, PowerPoint, Zotero

Honors and Awards

• University Merit Scholarship (awarded in all 8 semesters) • University Dean’s List Scholarship (awarded twice, during freshman and sophomore years) • Education Board Technical Scholarship, BUET • First Prize in the “NAME Olympiad” at NAME Fest 2023 • Champion and recipient of \$5,000 in prize money, Worldwide Ferry Safety Association Design Competition 2025, Team Black Pearl • Second Runner-up and recipient of \$1,000 in prize money, Worldwide Ferry Safety Association Design Competition 2026, Team Black Pearl • Award and certificate from the Chairman of BIWTC in recognition of outstanding performance in the Worldwide Ferry Safety Association Design Competition	
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Extracurricular Activities

• Class Representative , Department of Naval Architecture and Marine Engineering, BUET	<i>2022 – 2023</i>
• Volunteer , MARTEC 2022 – International Conference on Marine Technology	<i>Dec. 2022</i>
• Team Member, Black Pearl, BUET , multidisciplinary student team competing in the Worldwide Ferry Safety Association Design Competition	<i>Sept. 2024 – Present</i>
• Vice President , BUET Photographic Society	<i>Jun. 2025 – Jul. 2026</i>
• Director of Design and Creation , BUET Entrepreneurship Development Club	<i>Jun. 2025 – Jul. 2026</i>

References

Dr. Mir Tareque Ali Professor Department of Naval Architecture and Marine Engineering, BUET Contact: mtarequeali@name.buet.ac.bd Relationship: Junior Year Project Supervisor	Dr. Md. Shahidul Islam Professor Department of Naval Architecture and Marine Engineering, BUET Contact: shahid777@name.buet.ac.bd Relationship: Undergraduate Thesis Supervisor
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